

Intersections of Graphs

The intersections of the graphs of $f(x)$ and $g(x)$ can help you determine the x -values where $f(x) = g(x)$.

Graphically Solving $f(x) = g(x)$

When two functions are graphed on the same set of axes, either they will not intersect or they will intersect at one or more points. The points of intersection determine the x -values for which the functions are equal.

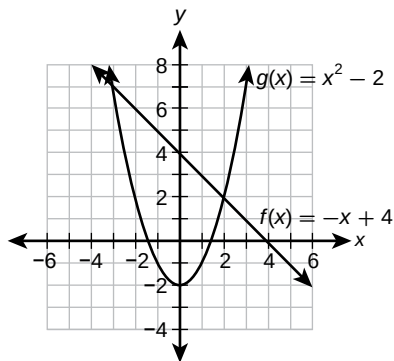
► **Think About It** If the graphs of two functions don't intersect, then there is no x -value where the functions are equal.

EXAMPLE 1

Determine the solutions to $f(x) = g(x)$, where $f(x) = -x + 4$ and $g(x) = x^2 - 2$.

SOLUTION

Graph $f(x)$ and $g(x)$ on the same set of axes.



By inspection, it appears that the two graphs intersect at the points $(2, 2)$ and $(-3, 7)$. Substitute the x -value for each point into the functions to confirm that both $f(x)$ and $g(x)$ have the same values.

For $x = 2$:

$$\begin{array}{ll} f(x) = -x + 4 & g(x) = x^2 - 2 \\ f(2) = -(2) + 4 & g(2) = (2)^2 - 2 \\ f(2) = 2 & g(2) = 4 - 2 = 2 \end{array}$$

For $x = -3$:

$$\begin{array}{ll} f(x) = -x + 4 & g(x) = x^2 - 2 \\ f(-3) = -(-3) + 4 & g(-3) = (-3)^2 - 2 \\ f(-3) = 3 + 4 = 7 & g(-3) = 9 - 2 = 7 \end{array}$$

The functions have the same values at $x = 2$ and $x = -3$. The solutions to $f(x) = g(x)$ are $x = 2$ and $x = -3$. ■

Using Technology to Find Intersections of Graphs

Suppose you're not able to determine the x -values at which two functions intersect by inspection. You can use technology to graph the functions and use trace and zoom features to approximate their intersection.

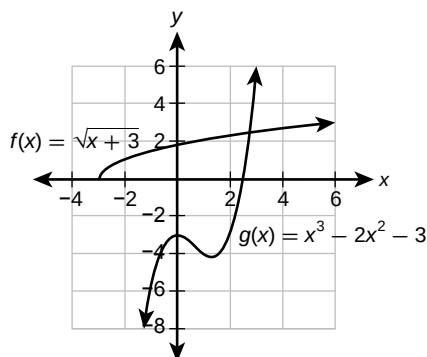
EXAMPLE 2

Determine the solutions to $f(x) = g(x)$: $\begin{cases} f(x) = \sqrt{x+3} \\ g(x) = x^3 - 2x^2 - 3 \end{cases}$.

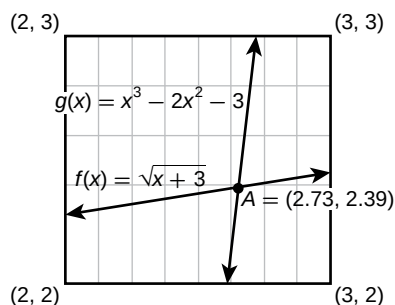
Round your answer to the nearest hundredth.

SOLUTION

Graph $f(x)$ and $g(x)$ on the same set of axes.



By inspection, it appears that the two graphs intersect when x is at slightly greater than 2. You can use technology (such as a zoom feature or setting a specific window on a graphing calculator) to get a closer look. This graph shows $f(x)$ and $g(x)$ with viewing window $2 < x < 3$ and $2 < y < 3$.



You can use the trace feature of your graphing technology to approximate the point of intersection. The graph shows the tracing of $f(x)$. The labeled point, which is very close to the point of intersection, shows a value of 2.73 for x where $f(x) = g(x)$. ■

- **Think About It** Some CAS will automatically determine the best approximation of the coordinates of the points of intersection of two graphed objects.

Solving Problems Involving Intersections of Graphs

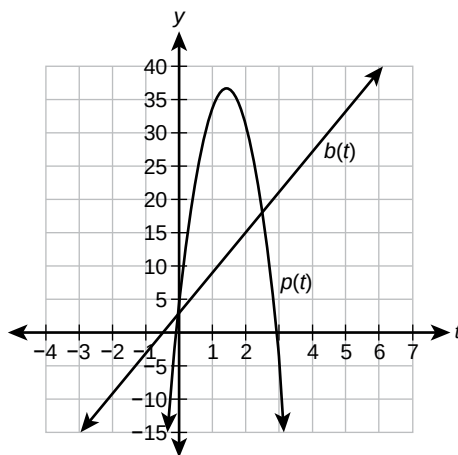
You can use graphing technology to solve problems involving the intersection of graphs.

EXAMPLE 3

After t seconds, the height, p , of a projectile (in feet) is given by the equation $p(t) = -16t^2 + 45t + 5$. After t seconds, the height, b , of a balloon (in feet) is given by the equation $b(t) = 6t + 3$. The projectile and the balloon are launched at the same time. After how many seconds (to the nearest hundredth) are the projectile and the balloon at the same height?

SOLUTION

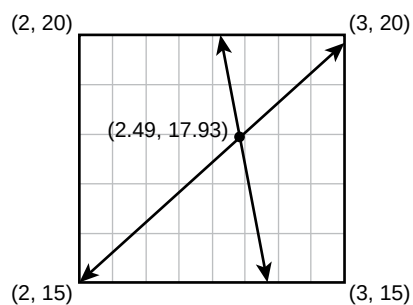
Graph $p(t) = -16t^2 + 45t + 5$ and $b(t) = 6t + 3$ on the same set of axes.



By inspection, it appears that the two graphs intersect when t is slightly less than zero and slightly greater than 2. Disregard the negative value of t for this problem, because time cannot be negative.

► **Think About It** The context of certain applications will eliminate some numbers as possible answers.

Use technology to confirm the point of intersection for the positive value of t . Set a specific window so that you can look at the graph closely. This graph shows viewing window $2 < t < 3$ and $15 < y < 20$.



The labeled point is the approximate point of intersection with a value of 2.49 for the value of t . The projectile and the balloon will be the same height after about 2.49 s. ■